DDR

FIXED_IO

TMDS_out

axi_bram_ctrl_vision

AXI BRAM Controller

S_AXI s_axi_aclk BRAM_PORTA
s_axi_aresetn

axi_dynclk

Dynamic Clock Generator (Pre-Production)

S_AXI_LITE REF_CLK_I PXL_CLK_O

PXL_CLK_5X_O
LOCKED_O

s_axi_lite_aclk
s_axi_lite_aresetn

axi_gpio_probe0

AXI GPIO

S_AXI GPIO

gpio_io_i[31:0] GPIO2
gpio2_io_i[31:0]

s_axi_aclk
s_axi_aresetn

axi_gpio_probe1

AXI GPIO

S_AXI GPIO

gpio_io_i[31:0] GPIO2
gpio2_io_i[31:0]

s_axi_aclk
s_axi_aresetn

axi_gpio_probe2

AXI GPIO

S_AXI GPIO

gpio_io_i[31:0] GPIO2
gpio2_io_i[31:0]

s_axi_aclk
s_axi_aresetn

axi_gpio_probe3

AXI GPIO

S_AXI GPIO

gpio_io_o[31:0] GPIO2
gpio2_io_i[31:0]

s_axi_aclk
s_axi_aresetn

axi_gpio_probe4

AXI GPIO

S_AXI GPIO

gpio_io_i[31:0] GPIO2
gpio2_io_i[31:0]

s_axi_aclk
s_axi_aresetn

axi_gpio_probe5

AXI GPIO

S_AXI GPIO

gpio_io_i[31:0] GPIO2
gpio2_io_i[31:0]

s_axi_aclk
s_axi_aresetn

axi_gpio_probe6

AXI GPIO

S_AXI GPIO
gpio_io_i[31:0] s_axi_aclk
s_axi_aresetn

axi_gpio_vision

AXI GPIO

S_AXI GPIO

gpio_io_o[31:0] GPIO2
gpio2_io_i[31:0]

s_axi_aclk
s_axi_aresetn

axi_gpio_vstat

AXI GPIO

S_AXI GPIO

gpio_io_i[31:0] GPIO2
gpio2_io_i[31:0]

s_axi_aclk
s_axi_aresetn

axi_mem_intercon

AXI Interconnect

S00_AXI

M00_AXI

S01_AXI S02_AXI S03_AXI ACLK

ARESETN
S00_ACLK

S00_ARESETN M00_ACLK M00_ARESETN

S01_ACLK
S01_ARESETN
S02_ACLK
S02_ARESETN
S03_ACLK
S03_ARESETN

axi_mem_intercon_out

AXI Interconnect

S00_AXI

M00_AXI

ACLK
ARESETN
S00_ACLK

S00_ARESETN M00_ACLK M00_ARESETN

axi_vdma_out

AXI Video Direct Memory Access

S_AXI_LITE M_AXI_MM2S M_AXIS_MM2S
s_axi_lite_aclk m_axi_mm2s_aclk m_axis_mm2s_aclk axi_resetn mm2s_frame_ptr_in[5:0]

mm2s_frame_ptr_out[5:0] mm2s_introut

axi_vdma_0

AXI Video Direct Memory Access

S_AXI_LITE

M_AXI_S2MM

S_AXIS_S2MM

s_axis_s2mm_tkeep[0:0] s_axi_lite_aclk m_axi_s2mm_aclk
s_axis_s2mm_aclk axi_resetn
s2mm_frame_ptr_in[5:0]

s2mm_frame_ptr_out[5:0] s2mm_introut

axis_probe_0

axis_probe_v1_0

m_axis

s_axis aclk aresetn

br_ovf br_unf br_sof_armed

br_dropping

br_wr_rst_busy br_rst_pclk

w0[31:0] w1[31:0] w2[31:0] w3[31:0]

blk_mem_gen_vision

Block Memory Generator

BRAM_PORTA

clka BRAM_PORTB

addrb[31:0] clkb

dinb[31:0] enb web[3:0]

clk_wiz_0

Clocking Wizard

clk_in1

clk_out1
clk_out2
locked

dvp_ctrl_0

dvp_ctrl_v1_0

axi_clk
clk24

ctrl[31:0] pl_pwdn
locked

xclk_out sensor_pwdn
vid_rst_force

rst_n_out dbg_trig
dbg_sel[2:0] dvp_probe_0

dvp_probe_v1_0

axi_clk pclk

vsync href scl sda
data[7:0] xclk pwdn

rst_n
locked

w0[31:0] w1[31:0] w2[31:0] w3[31:0] w4[31:0] w5[31:0] w6[31:0]

dvp2axis_bridge_0

dvp2axis_bridge_v1_0

m_axis pclk

rst_pclk href vsync data[7:0] aclk aresetn

ovf_sticky unf_sticky

sof_armed_obs dropping_obs wr_rst_busy_obs
rst_obs dbg_trig
dbg_sel[2:0] dbg_data[31:0]

frame_pump_0

frame_pump_v1_0

s_axi m_rd m_wr clk

rstn

hdmi_out_hpd[0:0]

hdmi_out_hpd_video

AXI GPIO

S_AXI GPIO
gpio_io_o[0:0] s_axi_aclk

s_axi_aresetn

line_engine_top_0

line_engine_top_v1_0

s_axi s_axi_aclk m_axi s_axi_aresetn

map_detect_axi_0

map_detect_axi_v1_0

s_axi s_axi_aclk
s_axi_aresetn
tap_wr_en
tap_wr_addr[14:0]
tap_wr_data[31:0]

dbg_frame_seq[31:0]

ov5640_data[7:0]

ov5640_top_0

ov5640_top_v1_0

s_rst_n
clk_sys50m
clk_sys24m

ov5640_pwdn
ov5640_rst_n

ov5640_xclk ov5640_iic_scl ov5640_iic_sda
ov5640_pclk ov5640_href ov5640_vsync ov5640_data[7:0] m_data[15:0] m_wr_en

estart ewdata[31:0] riic_data[7:0]

pe_axi_0

pe_axi_v1_0

s_axi s_axi_aclk
s_axi_aresetn

led_locked

ov5640_href

ov5640_iic_scl ov5640_iic_sda

ov5640_pclk

ov5640_pwdn

ov5640_rst_n

ov5640_vsync

ov5640_xclk

processing_system7_0

ZYNQ7 Processing System

DDR
FIXED_IO
USBIND_0

S_AXI_HP0_FIFO_CTRL S_AXI_HP1_FIFO_CTRL

M_AXI_GP0
S_AXI_HP0 S_AXI_HP1 M_AXI_GP0_ACLK

S_AXI_HP0_ACLK
S_AXI_HP1_ACLK

FCLK_CLK0 FCLK_RESET0_N

ps7_0_axi_periph

AXI Interconnect

S00_AXI

M00_AXI M01_AXI M02_AXI M03_AXI M04_AXI M05_AXI M06_AXI M07_AXI M08_AXI M09_AXI M10_AXI M11_AXI M12_AXI M13_AXI M14_AXI M15_AXI M16_AXI M17_AXI M18_AXI M19_AXI

ACLK
ARESETN
S00_ACLK

S00_ARESETN M00_ACLK M00_ARESETN M01_ACLK M01_ARESETN M02_ACLK M02_ARESETN M03_ACLK M03_ARESETN M04_ACLK M04_ARESETN M05_ACLK M05_ARESETN M06_ACLK M06_ARESETN M07_ACLK M07_ARESETN M08_ACLK M08_ARESETN M09_ACLK M09_ARESETN M10_ACLK M10_ARESETN M11_ACLK M11_ARESETN M12_ACLK M12_ARESETN M13_ACLK M13_ARESETN M14_ACLK M14_ARESETN M15_ACLK M15_ARESETN M16_ACLK M16_ARESETN M17_ACLK M17_ARESETN M18_ACLK M18_ARESETN M19_ACLK M19_ARESETN

rgb2dvi_0

RGB to DVI Video Encoder (Source) (Pre-Production)

TMDS

RGB

aRst_n PixelClk SerialClk rst_pixel

Processor System Reset

slowest_sync_clk ext_reset_in

aux_reset_in mb_debug_sys_rst dcm_locked

mb_reset bus_struct_reset[0:0] peripheral_reset[0:0]
interconnect_aresetn[0:0] peripheral_aresetn[0:0]

rst_ps7_0_100M

Processor System Reset

slowest_sync_clk ext_reset_in

aux_reset_in mb_debug_sys_rst dcm_locked

mb_reset bus_struct_reset[0:0] peripheral_reset[0:0]
interconnect_aresetn[0:0] peripheral_aresetn[0:0]

util_vector_logic_0

Utility Vector Logic

Op1[0:0] Res[0:0]

v_axi4s_vid_out_0

AXI4-Stream to Video Out

video_in
vtiming_in

vid_io_out

aclk aclken
aresetn
fid
vid_io_out_clk
vid_io_out_ce
vid_io_out_reset

vtg_ce
locked
overflow
underflow

fifo_read_level[11:0] status[31:0] sof_state_out

vid_rst_ctrl_0

vid_rst_ctrl_v1_0

pclk
locked_n
force_rst

vid_rst

vision_core_0

vision_core_v1_0

m_axis

s_axis aclk aresetn

freeze

soft_rst swap_bytes

frame_cnt[31:0] stat[31:0] stat2[31:0] wr_en wr_addr[14:0] wr_data[31:0]

vtc_out

Video Timing Controller

ctrl

vtiming_out

clk
clken
s_axi_aclk
s_axi_aclken
gen_clken
sof_state
resetn
s_axi_aresetn

irq

fsync_in

fsync_out[0:0]

v3_bundle_axi_0

v3_bundle_axi_v1_0

s_axi s_axi_aclk
s_axi_aresetn

xl_hdmi_ce

Constant

dout[0:0]

xl_hdmi_one

Constant

dout[0:0]

xl_hdmi_zero

Constant

dout[0:0]

xlconstant_v_enb

Constant

dout[0:0]

xlconstant_1

Constant

dout[0:0]

xlconstant_2

Constant

dout[0:0]

xlconstant_3

Constant

dout[31:0]

xlslice_v_freeze

Slice

Din[31:0] Dout[0:0]

xlslice_v_srst

Slice

Din[31:0] Dout[0:0]

xlslice_v_swap

Slice

Din[31:0] Dout[0:0]